

Digital Image Processing: 3-0-2 (42 Lectures)

Module	Topics	Reference (Gonzalez 4th Ed.)	Lectures
Module 1	Introduction to Digital Image Processing: Applications, DIP vs Computer Vision, Image Acquisition, Image Sensors, Human Visual System, Image Formation, Pixels, Resolution, Sampling, Quantization, Image Representation	Ch. 1–2	4
Module 2	Image Enhancement in Spatial Domain: Intensity Transformations, Histogram Processing, Contrast Stretching, Histogram Equalization, Log/Gamma Transformations, Image Arithmetic	Ch. 3	4
Module 3	Spatial Filtering: Convolution, Smoothing Filters, Sharpening Filters, Noise Models, Median, Gaussian, Laplacian, High-Boost Filtering	Ch. 3	5
Module 4	Frequency Domain Processing: Fourier Transform, DFT, FFT, Frequency Domain Enhancement, Ideal, Butterworth and Gaussian Filters, Introduction to Wavelets	Ch. 4	5
Module 5	Color Image Processing & Morphology: Color Models (RGB, HSV, Lab), Pseudocolor, Dilation, Erosion, Opening, Closing, Boundary Extraction	Ch. 6 & Ch. 9	4
Total Conventional Digital Image Processing			22

Modern Image Processing -Analysis and Computer Vision (20 Lectures)

Module	Topics	Reference	Lectures
Module 6	Feature Extraction & Image Representation: Edge Detection (Sobel, Prewitt, Canny), Corners (Harris), Texture Features, Shape Features, Feature Descriptors (SIFT, SURF, ORB), Feature Matching	Gonzalez Ch. 10–11, Szeliski	4
Module 7	Image Segmentation & Pattern Classification: Thresholding, Otsu, Region Growing,	Gonzalez Ch. 10–11	4

	Watershed, K-Means Segmentation, Image Classification, Feature Selection		
Module 8	Machine Learning for Image Analysis: Supervised vs Unsupervised Learning, KNN, Decision Trees, SVM, Random Forest, Naïve Bayes, Clustering for Image Classification	Szeliski, ML references	4
Module 9	Deep Learning & Modern Computer Vision: CNN Fundamentals, Transfer Learning, Image Classification, Object Detection (YOLO), Semantic Segmentation (U-Net), Instance Segmentation, Image Captioning	Goodfellow, YOLO, U-Net	4
Module 10	Video Processing & Vision Transformers: Video Representation, Motion Detection, Optical Flow, Object Tracking, Action Recognition, Vision Transformers (ViT), DETR, Segment Anything (SAM), Foundation Vision Models, Multimodal Vision	ViT, DETR, SAM, CLIP	4

Lab No.	Experiment	Platform
1	Image acquisition, display, sampling & quantization	MATLAB
2	Histogram processing & intensity transformations	MATLAB
3	Spatial filtering (smoothing & sharpening)	MATLAB
4	Noise generation & image restoration	MATLAB
5	Frequency-domain filtering using FFT	MATLAB
6	Color image processing	MATLAB
7	Morphological operations	MATLAB
8	Edge detection & feature extraction	MATLAB
9	Image segmentation	MATLAB
10	Image classification using classical ML	MATLAB
11	CNN-based image classification & YOLO demonstration	Python (PyTorch/OpenCV)
12	Vision Transformer / Video Processing demonstration	Python

A. Primary Textbook (Mandatory)

1. [Digital Image Processing \(4th Edition\) – Gonzalez & Woods \(Pearson\)](#)

Authors

- Rafael C. Gonzalez
- Richard E. Woods

This should be the **primary textbook** for the course.

Relevant chapters:

Module	Chapters
Introduction	Ch. 1
Image Fundamentals	Ch. 2
Enhancement	Ch. 3
Frequency Domain	Ch. 4
Restoration	Ch. 5
Wavelets	Ch. 6
Color Processing	Ch. 7
Compression	Ch. 8 (selected)
Morphology	Ch. 9
Segmentation	Ch. 10
Feature Extraction & Recognition	Ch. 11–12

The 4th edition also includes expanded material on **deep learning, SIFT, graph cuts, superpixels, active contours, and 120 MATLAB projects**, making it suitable for a modern DIP course.

B. MATLAB Companion (Highly Recommended)

2. [Digital Image Processing Using MATLAB \(3rd Edition\)](#)

Authors

- Gonzalez
- Woods
- Steven Eddins (MathWorks)

This book complements the primary text by providing:

- MATLAB implementations
- Image Processing Toolbox usage
- Deep Learning Toolbox examples
- More than 130 programming projects
- Downloadable support packages and image datasets

C. Companion Computer Vision Text

3. [Computer Vision: Algorithms and Applications \(Richard Szeliski\)](#)

Recommended chapters:

- Feature Detection
- Stereo Vision

- Motion
- Recognition
- Segmentation
- Deep Learning

This is one of the best free references for bridging classical DIP with modern Computer Vision. It is widely recommended alongside Gonzalez & Woods.

D. MATLAB Documentation

4. [MATLAB Image Processing Toolbox Documentation](#)

Useful for:

- `imread`
- `imshow`
- Histogram processing
- Filtering
- FFT
- Morphology
- Segmentation
- Feature extraction

5. [MATLAB Deep Learning Toolbox Documentation](#)

Useful for:

- CNNs
- Transfer Learning
- Object Detection
- Semantic Segmentation
- Vision Transformers (selected examples)

E. OpenCV

6. [OpenCV Documentation](#)

Reference for

- Image I/O
- Filtering
- Feature Detection
- Edge Detection
- Morphology
- Segmentation
- Tracking
- Object Detection

F. Python Image Processing

7. [scikit-image Documentation](#)

Excellent for

- Thresholding
- Morphology
- Filtering
- Segmentation
- Region Properties
- Feature Extraction

The library is widely used in research and education for image processing.

8. [scikit-learn Documentation](#)

Recommended for

- KNN
- SVM
- Random Forest
- Decision Trees
- Clustering
- PCA

G. NPTEL Courses

9. [NPTEL – Digital Image Processing](#)

Recommended for

- Theory
- Mathematical derivations
- Assignments

10. [NPTEL – Computer Vision](#)

Recommended for

- Camera Models
- Feature Detection
- Recognition
- Geometry
- Tracking

H. Deep Learning Resources

11. [PyTorch Tutorials](#)

Useful for

- CNN
- Transfer Learning
- Image Classification
- Object Detection

12. [Ultralytics YOLO Documentation](#)

Recommended for

- YOLO
- Object Detection
- Segmentation
- Tracking

13. [Hugging Face Transformers Documentation](#)

Recommended for

- Vision Transformer (ViT)
- DETR
- CLIP
- Multimodal Vision

I. MATLAB-Based Laboratory References

Lab	Reference
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Image Reading	MATLAB Image Processing Toolbox
Histogram	Gonzalez Ch. 3 + MATLAB
Enhancement	Gonzalez Ch. 3
Filtering	MATLAB + Gonzalez
FFT	Gonzalez Ch. 4
Frequency Filtering	MATLAB
Color Processing	Gonzalez Ch. 7
Morphology	Gonzalez Ch. 9
Segmentation	Gonzalez Ch. 10
Feature Extraction	Szeliski
CNN Demonstration	MATLAB Deep Learning Toolbox
YOLO / ViT Demonstration	PyTorch + Ultralytics + Hugging Face

Suggested Reference List for the Syllabus (Download the coloured references)

1. **R. C. Gonzalez and R. E. Woods**, *Digital Image Processing*, 4th Edition, Pearson. (Primary Text)
2. **R. C. Gonzalez, R. E. Woods, and S. L. Eddins**, *Digital Image Processing Using MATLAB*, 3rd Edition.
3. **Richard Szeliski**, *Computer Vision: Algorithms and Applications* (Free online).
4. **Wilhelm Burger and Mark Burge**, *Digital Image Processing: An Algorithmic Introduction using Java*.
5. **Anil K. Jain**, *Fundamentals of Digital Image Processing*.
6. **OpenCV Documentation**.
7. **MATLAB Image Processing Toolbox Documentation**.
8. **scikit-image Documentation**.
9. **PyTorch Tutorials**.
10. **Ultralytics YOLO Documentation**.

This combination gives students a complete learning path:

- **Gonzalez & Woods** for theory and mathematics,
- **Digital Image Processing Using MATLAB** for laboratory implementation,
- **Szeliski** for modern computer vision concepts,
- **PyTorch/YOLO/Hugging Face** for the final deep learning and transformer modules.